

Here's one more pair of results to justify why we will care about meromorphic functions a lot once we define them in a second.

Maximum Modulus Theorem: Suppose M is a connected Riemann surface and $F : M \rightarrow \mathbb{C}$ is a holomorphic function. If $|F|$ attains a local maximum on M then F is a constant function.

Proof:

Suppose $|F|$ has a local maximum at $p \in M$. Then let (U, φ) be a complex chart containing p . By shrinking U if necessary, we can also without loss of generality assume $\varphi(U)$ is a connected set. But now $F \circ \varphi^{-1}$ is a holomorphic function on the region $\varphi(U)$ of $\mathbb{C}$ which also satisfies that $|F \circ \varphi^{-1}|$ attains a local maximum at $\varphi(p)$. It follows by the maximum modulus principle from complex analysis (see [page 384](#)) that $F \circ \varphi^{-1}$ is a constant function on $\varphi(U)$. In turn $F \circ \varphi^{-1} \circ \varphi = F$ is also a constant function on U . And by the identity principle, we can then conclude that F is constant on M . ■

As a side note since I realize that I never showed this, any constant function between Riemann surfaces is automatically holomorphic. To see why, suppose M and N are Riemann surfaces, and $F : M \rightarrow N$ satisfies that $F(p) = q \in N$ for all $p \in M$. Then having picked a complex chart (V, ψ) in N containing Q , we have for every complex chart (U, φ) in M that $F(U) \subseteq V$ and $\psi \circ F \circ \varphi^{-1}$ is a constant function from $\varphi(U) \subseteq \mathbb{C}$ into $\mathbb{C}$. So, $\psi \circ F \circ \varphi^{-1}$ is trivially holomorphic.

Corollary: If M is a compact Riemann surface then $O(M, \mathbb{C})$ consists entirely of locally constant functions. In particular, if M is connected then $O(M, \mathbb{C})$ consists only of constant functions.

Proof:

Note that every connected component of M is closed. And since closed subsets of compact sets are compact, we can thus conclude that every connected component of M is compact.

Also note that since M is locally connected, we know that every connected component of M is open. Therefore every connected component of M (equipped with the subspace topology) can still be viewed as a Riemann surface where F restricted to each component is holomorphic.

Finally, suppose $F \in O(M, \mathbb{C})$ and let N be any connected component of M . Then F being continuous on N plus N being compact implies that $|F|$ attains a local maximum on N . Yet also note that F being holomorphic on N as a Riemann surface means that the maximum modulus theorem applies. So, we must have that F is constant on N . ■

Since a lot of the Riemann surfaces we actually care about are compact and connected, we thus can see that $O(M, \mathbb{C})$ is often not the collection of functions we want to study on a Riemann surface. Hence, we will instead often consider $O(M, \mathbb{C}_\infty)$.

To motivate the next bit, observe that if $U \subseteq \mathbb{C}$ is an open set then any meromorphic function $f : U \rightarrow \mathbb{C}$ extends to a holomorphic function from U into $\mathbb{C}_\infty$ (in the Riemann surface sense).

If you squint you can see that this is just my comment from [page 467](#). After all, $z \mapsto f(z)$ is a holomorphic function from $U - f^{-1}(\{\infty\})$ into $\mathbb{C}$ and $z \mapsto 1/f(z)$ is a holomorphic function from $U - f^{-1}(\{0\})$ into $\mathbb{C}$.

Conversely, suppose $G \subseteq \mathbb{C}$ is a region and $f : G \rightarrow \mathbb{C}_\infty$ is a holomorphic function (in the Riemann surface sense) satisfying that there exists $z \in G$ with $f(z) \neq \infty$. Since f is not the constant ∞ -valued function, we know by the identity principle that $f^{-1}(\{\infty\})$ can't have any limit points. This also means every $a \in f^{-1}(\{\infty\})$ must be an isolated point of G .

Let $\Omega := G - f^{-1}(\{\infty\})$. Since f is continuous, we can conclude that Ω is an open subset of G . So, we consider the chart $(\Omega, \text{Id}_\Omega)$. Next, we consider the chart $(\mathbb{C}, \text{Id}_\mathbb{C})$ in the standard structure on $\mathbb{C}_\infty$. So, $\text{Id}_\mathbb{C} \circ f \circ \text{Id}_\Omega = f$ is a holomorphic function from Ω into $\mathbb{C}$ (in the complex analytic sense).

But now $f|_\Omega$ has singularities at every $a \in f^{-1}(\{\infty\})$. Also by the continuity of f on G , we know that $|f(z)| \rightarrow \infty$ as $z \rightarrow a$. Therefore, we can conclude that $f|_\Omega$ has poles at all $a \in G - \Omega$. Or in other words, f is a meromorphic function on G (in the complex analytic sense).

All in all, we can conclude that if $U \subseteq \mathbb{C}$ is an open set, then the set of meromorphic functions on U is precisely equal to $O(U, \mathbb{C}_\infty) - \mathcal{F}$ where $\mathcal{F}$ is the collection of holomorphic functions that are constantly ∞ on any neighborhood. For the time being, I'm going to denote the set of meromorphic functions on U as $\mathcal{M}(U)$.

I realize that $\mathcal{M}$ and M look similar. So, I'll probably start denoting Riemann surfaces by X and Y instead.

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